

Local Subsidies Fraud Risk - Model Comparison

Note: Korean narrative is in docs/metrics_guide.md. This PDF summarizes aggregate metrics only (no raw rows).

Train period: 202401-202506

Test period: 202507-202512

Risk score range: 0-1000 (higher = higher fraud risk).

Reports contain aggregate statistics only. No raw PII rows.

See docs/metrics_guide.md for Korean explanations of metrics.

Per-algorithm reports: outputs/reports/{algorithm}/

Model metrics (aggregate)

algorithm=catboost,	Accuracy=0.9906432546403512,	Precision=0.21364985163204747,	Recall=0.9266409266409267,
F1=0.3472389679286231,	ROC_AUC=0.9762611907184625		
algorithm=stacked_ensemble,	Accuracy=0.9364349659534755,	Precision=0.03837081302091524,	Recall=0.9420849420849421,
F1=0.07373828951344817,	ROC_AUC=0.9783329722782055		
algorithm=easy_ensemble,	Accuracy=0.7242611731360824,	Precision=0.007444725717349827,	Recall=0.7683397683397684,
F1=0.014746566544807825,	ROC_AUC=0.8378898048866475		
algorithm=gradient_boosting,	Accuracy=0.9868929521966057,	Precision=0.1576198046786282,	Recall=0.8931788931788932,
F1=0.26795366795366793,	ROC_AUC=0.9769659117168761		
algorithm=random_forest,	Accuracy=0.9983927275241091,	Precision=0.6793103448275862,	Recall=0.7606177606177607,
F1=0.7176684881602914,	ROC_AUC=0.9845367865663638		